AIAA Meeting Papers on Disc, January 1997
A9715529, NAS2-14031, AIAA Paper 97-0471

Composite heat shields revisited

Gary A. Allen, Jr.
NASA, Ames Research Center, Moffett Field, CA

AIAA, Aerospace Sciences Meeting & Exhibit, 35th, Reno, NV, Jan. 6-9, 1997

Composite heat shields are multistaged heat shields. The first stage is used to dissipate kinetic energy during atmospheric entry. The first stage is then jettisoned to expose a second stage optimized for maneuverability. A history is provided of the composite heat shield describing the original concept synthesis and practical problems encountered. A new practical design for a composite heat shield using flight-proven hardware is described. To investigate this new design, an atmospheric entry simulation program was written and validated using Viking-I flight data. The atmospheric entry program demonstrated that the new composite heat shield using conventional thermal protection materials is practical for Martian atmospheric entry. Different potential applications for a composite heat shielded spacecraft are considered, including a low-cost Mars sample return mission. (Author)

Composite Heat Shields Revisited

Gary A. Allen, Jr.*

Thermosciences Institute, NASA Ames Research Center, Moffett Field, CA

Abstract

Composite heat shields are multi-staged heat shields. The first stage is used to dissipate kinetic energy during atmospheric entry. The first stage is then jettisoned to expose a second stage optimized for maneuverability. A history is provided of the composite heat shield describing the original concept synthesis and practical problems encountered. A new practical design for a composite heat shield using flight proven hardware is described. To investigate this new design, an atmospheric entry simulation program was written and validated using Viking-I flight data. The atmospheric entry program demonstrated that the new composite heat shield using conventional thermal protection materials is practical for Martian atmospheric entry. Different potential applications for a composite heat shielded spacecraft are considered including a low-cost Mars sample return mission.

Introduction

Precision landing onto the surface of Mars imposes many conflicting constraints on a spacecraft thermal protection system. The concept of using aerocapture or ballistic entry to bring a spacecraft from heliocentric orbit to Mars orbit or the Martian surface has been studied for decades.[1-3] These studies have shown there can be a significant improvement in spacecraft weight reduction and overall mission flexibility if the atmosphere of Mars is used to reduce spacecraft orbital velocity rather than on-board propellant.

Many heliocentric trajectories require a high approach velocity to Mars. Manned Mars missions in particular dictate high energy orbits to minimize time in deep space in order to reduce crew exposure to radiation and zero gravity. Therefore direct Martian atmospheric entry from heliocentric orbit can lead to a significant thermal load on the heat shield.

*Senior Research Scientist, Member AIAA.

Allen and Eggers classic paper on blunt body heat shields showed that changing a heat shield's shape to increase drag reduced the heat shield's total heat load thus allowing for weight reduction in the heat shield's thermal protection mass.[4] However there are many reasons why a pure drag heat shield (without lift) is unacceptable for precision landing or orbit capture. For example, without lift the spacecraft would have no cross range capability and thus be unable to deflect its trajectory towards the target area. This can result in a landing error ellipse being on the order of a hundred kilometers as is the case with Mars Pathfinder's nonlifting entry. Employing lift (particularly negative lift) is an important technique for reducing peak deceleration loads during aerobraking. Mars Pathfinder will experience a maximum deceleration of almost 20 gravities[5] while the maximum tolerable for a manned mission is about five gravities.[6]

Experience has shown that enhancing the maximum lift-over-drag ratio [L/D] normally results in increasing the ballistic parameter [Mass/(C_d Area)] either by reducing the drag or increasing the spacecraft's mass.[7] Heat shield heat load increases with the ballistic parameter. Drag reduction is typically achieved by reducing the nose radius which in turn increases the stagnation point convective heat flux[8] as well as the total heat load due to convection as shown by Allen and Eggers. The reduction in nose radius can also lead to diminishing the heat shield volume available for payload. Thus there is a fundamental contradiction imposed by the physics of atmospheric entry, viz. for a light weight thermal protection system there must be high drag but for maneuverability there must be low drag.

Chapman's Composite Heat Shield

In 1958, Dean Chapman in his seminal paper[9] on atmospheric entry anticipated these conflicting requirements for precision atmospheric entry. To overcome these difficulties, Chapman devised a multistaged heat shield for use in what he called "composite entry". This composite entry heat shield or *composite heat shield* in its simplest form was a two staged heat shield. The first stage was a pure drag heat shield optimized to reduce vehicle velocity. This first stage was jettisoned once the vehicle's velocity diminished to an acceptable magnitude. After jettisoning the first stage heat shield, a second stage heat shield was exposed. This second stage heat shield had a lifting geometry and provided the necessary cross range capability for accurate landing. Chapman's first report was favorable towards this concept. However in a subsequent report[10] Chapman showed that for reentry into the Earth's atmosphere from lunar or heliocentric orbit, the spacecraft's trajectory would first pass through the atmosphere using the first stage heat shield and then enter into a suborbital trajectory. This suborbital trajectory would then spend a lengthy time in the Van Allen belts exposing its payload to harmful radiation before returning to the atmosphere for final entry using the second stage heat shield. It was because of the issue of radiation that the composite heat shield was originally considered impractical and not pursued.

Composite heat shields are indeed impractical for planets with radiation belts. Unfortunately this was not the only problem. In his subsequent report Chapman also showed an error in the original analysis. Chapman earlier assumed the angle from the planet's local horizon to the spacecraft's velocity vector (gamma) could be determined with perfect accuracy. In actual practice this angle is calculated on Earth using data from the Deep Space Network in conjunction with data acquired from the spacecraft using inertial measurements and optical sensors, i.e. a star tracker. Data of this sort always has some error that propagates to the calculated value of gamma. The effect of this error in gamma magnifies trajectory error when there is no L/D as is the case in unguided atmospheric entry. Consequently a pure drag heat shield can not provide accurate atmospheric entry as the first stage of a composite heat shield.

New Approach to the Composite Heat Shield

The two difficulties of the Earth's radiation belts and initial trajectory error lead Chapman not to pursue the original composite heat shield concept. However the defects of Chapman's original concept can be repaired.

The first defect is easily repaired: Do not use the composite heat shield for entry into the Earth's atmosphere. Instead use it for entry into the atmosphere of a planet without radiation belts, i.e. Mars or Venus.

Repair of the second defect requires inventing a specific composite heat shield designed for entry into the atmosphere of Mars. The first stage of this composite heat shield is a 70 degree half angle sphere-cone geometry based upon the Pathfinder aeroshell. Pathfinder is scheduled to land on Mars in July 1997. The Pathfinder aeroshell has a high drag, low lift geometry with a mass of 74 kg and complete vehicle ballistic coefficient of 62 kg/m^2. The Pathfinder aeroshell is based upon the Viking aeroshell that was twice successfully used for landing on Mars in 1976. For the second stage heat shield, the geometry of the Advanced Maneuverable Reentry Vehicle (AMaRV) is selected. AMaRV was a successfully tested weapon system prototype designed for rapid and accurate reentry into the Earth's atmosphere while performing high-G maneuvers. AMaRV had a biconic-cut body with a split windward flap and two side flaps for yaw control. The half angle of AMaRV's leading cone was 10.4 degrees and a trailing cone was 6 degrees.[11] AMaRV's low drag, relatively high lift geometry had a mass of 470 kg and a ballistic coefficient of 13485 kg/m^2. Though originally a classified program, some information pertaining to AMaRV has been declassified for supporting DC-X and other unclassified projects.

Integrated within the first stage Pathfinder aeroshell is an instrument-controller ring with a center of mass coincident with the aeroshell's center of mass. This instrument-controller ring is given a mass of 327 kg so as to set the total vehicle center of mass to the same location as the actual Pathfinder spacecraft's prior to Martian atmospheric entry. This total vehicle shall be called the "Composite Heat Shielded Spacecraft" (CHeSS). Fig. 1 shows a sketch of the CHeSS prior to initial atmospheric entry and the separated AMaRV second stage. The AMaRV sketch is correctly scaled in size to the Pathfinder aeroshell but not accurate in detail.

Pathfinder carries a "backpack" during its transfer from Earth to Mars. The backpack weighs 320 kg and provides solar power, on-orbit navigation, etc. The backpack is jettisoned prior to atmospheric entry. It is assumed the CHeSS instrument-controller ring performs the same functions as the Pathfinder backpack. Neglecting reaction-control propellant loss, the launch and entry mass of CHeSS is assumed to be 871 kg which is less than Pathfinder's launch mass of 890 kg. Assuming a drag coefficient of 1.6, the ballistic coefficient for CHeSS is 98.6 kg/m^2. CHeSS could fit within the same volume of the Delta-II launch faring used by Pathfinder.

CHeSS was simulated to enter the atmosphere of a rotating Mars at a nominal inertial entry angle of -12.5 degrees with an angle-of-attack of -11.1 degrees (Viking's nominal angle of attack). The entry altitude of 125 km and corresponding inertial velocity of 7.5 km/sec for CHeSS were the same values planned for Pathfinder. However CHeSS was simulated to enter the Martian atmosphere in a west-to-east orbit rather than a retrograde orbit. Because the CHeSS performs multiple skips off of the Martian atmosphere there was a significant disadvantage to a retrograde orbit.

As was found by Chapman, a composite heat shield must have some lift for capture to an accurate orbit. The control strategy for this study is to have CHeSS skip off of the Martian atmosphere with its first stage to a suborbital trajectory. After leaving the Martian atmosphere, the CHeSS jettisons its first stage heat shield while experiencing no significant aerodynamic forces. After this point the CHeSS is aerodynamically identical to AMaRV, see Fig. 2. As an AMaRV the CHeSS has a nominal angle of attack of 10.0 degrees. This is in the middle of AMaRV's operating range. In this orientation the CHeSS skips off of the Martian atmosphere five more times before impacting the surface as shown in Fig. 3. The trajectory program that produced Fig. 3 is described in the next section of this paper. It is assumed the first stage heat shield removes large scale trajectory error while the second stage fine tunes the trajectory. It is required that the range of atmospheric skips by the AMaRV second stage overlap so that the second stage can impact to the same location on the surface of Mars. The requirement of skip range overlap forces the first skip as an AMaRV (the second skip for CHeSS) to be at almost the same location. If a fixed angle-of-attack without roll modulation for the first stage were assumed, CHeSS could only tolerate an error in gamma of less than plus-or-minus 0.1 degree which is not achievable using a conventional inertial navigation system. An error of plus-or-minus 0.5 degrees is typically assumed in mission planning.[12] Roll modulation could relieve this problem but there is the additional consideration of the second stage's geometry with respect to the first stage's wake bubble. For CHeSS to be controllable, the nose of the second stage must not touch the shear layer of the wake bubble. Therefore the angle of the second stage AMaRV's main axis with respect to the first stage heat shield's axis of symmetry must be variable. This could be achieved by using mechanical actuators much like gimballing a rocket nozzle, see Fig. 4. However given that the second stage must be gimballed with respect to the first stage then the most logical control method is not to use roll modulation but to vary the location of the center-of-mass. The trim of CHeSS would be set by having the aerodynamic moment, the moment caused by the first stage center-of-mass offset and the moment caused by the second stage center-of-mass offset summed to zero. Variation of the center-of-mass through this technique is called the *tail waging the dog* control strategy by Regan & Anandakrishnan[13]. It is assumed that by varying the location of the center of mass, the angle-of-attack for CHeSS can range from zero to -14 degrees with -11.1 degrees as the starting value. The use of the *tail waging the dog* control strategy repairs the second defect in Chapman's original concept.

Trajectory Simulation Method

The altitude versus time plot shown in Fig. 3 was produced using a new atmospheric entry simulation program named "Traj". Traj was written in the C programming language and integrates a three degree of freedom dynamical system. The equations of motion are in a vehicle centered form using theory described by N.V. Vinh.[14] The state variables are radius from the center of Mars, velocity magnitude, the angle of the velocity vector relative to the local horizon, latitude, longitude and heading angle. Angle of attack and roll angle are control parameters. The frame of reference was transformed back-and-forth from the inertial frame (nonrotating Mars) to the frame rotating with Mars. For the rotating frame, all centrifugal and Coriolis terms are included (exact equations of motion). The Mars atmospheric model was the mean value COSPAR model.[15] The MARSGRAM atmospheric model is popular with other investigators but carries a significant CPU time burden because it calculates mean temperature lapse rates. It was felt this additional CPU time burden did not sufficiently enhance simulation accuracy over the simpler COSPAR model. The Martian atmosphere was assumed to rotate as a solid body with the planet. Aerodynamic forces were always calculated. However during a simulation run when the aerodynamic forces were calculated to be less than 10^{-8} of the gravitational forces, the solution was transformed from a rotating frame to an inertial frame with the lift and drag components no longer included in the total force calculations. Since the AMaRV component of CHeSS skipped in and out of the atmosphere, the Traj program had to perform this transformation back-and-forth several times during a single run. Doing these transformations not only reduced the CPU time requirement but also avoided a numerical instability that occurred when the vehicle approached areosynchronous altitudes in a rotating frame of reference where the gravitational force magnitude equaled the Coriolis force magnitude.

The Mars gravitational model assumed an oblate spheroid with the J_2 harmonic of 0.001965 included using the A. Seiff[15] formulation. The convective heat flux was calculated using the Marvin-Deiwert[16] model for CO_2 with an assumed emissivity of 0.85. The radiative heat flux was calculated using the Tauber-Sutton[8] model. The radiative heat flux was never significant in this study but was validated during code development by simulating Martian entry for a large sized raked conic.

Quadratic spline fits for the lift and drag aerodynamic coefficients dependent upon the independent variables of Mach number and angle of attack were hard coded inside Traj for the Viking/Pathfinder and AMaRV geometries. Figures 5-8 showing the aerodynamic models for Viking/Pathfinder and AMaRV used for CHeSS were produced using those spline fits. The aerodynamic geometry could be changed during run-time thus permitting simulation of a composite heat shield. Beyond being a superb maneuvering vehicle, AMaRV was ideal for this study because actual flight test aerodynamic data was available. The test data ranged from Mach 3.5 to 22 for air. The atmosphere of Mars is mainly carbon dioxide which could cause the axial aerodynamic force coefficient to be 7% higher than the corresponding value for air.[17] However the Martian atmosphere is so tenuous that the local atmospheric density for a given altitude can vary over 7% depending upon the time of year or location on Mars.[15] Therefore the error introduced in using aerodynamics based upon air rather than carbon dioxide is usually less than the error in the Martian atmospheric model.

Traj incorporated within it two analytic models for code validation and error detection. This feature allows the user to validate the code for any given entry condition. The two analytic solutions are based upon either the no-gravity, simple isothermal atmospheric model or the no-atmosphere, dipole gravitational field model (conic section formula). The numerical integrator is a fourth-order Runge Kutta with a variable step size control algorithm (there are more efficient algorithms). The numerics were tuned against the analytic solutions. Traj was validated against Viking-I data. Fig. 9 shows the Viking-I entry trajectory produced by Traj compared to flight data from Viking's radar altimeter. For the actual Viking-I inertial entry angle of -16.99 deg., Traj calculated a peak heat flux of 20 watts/cm^2. Viking's manufacturer (Martin-Marietta) calculated 24 watts/cm^2 in their final report.[17] Traj calculated a peak deceleration of 70 meters/sec^2 which compares to 71 meters/sec^2 in the Martin-Marietta report. Traj calculated that Viking-I was at 22.8 deg. latitude and -47.4 deg. longitude when its parachutes were deployed. Viking-I spent 108 seconds from deploying it parachutes to actually landing on the Martian surface at 22.46 deg. latitude and -48.01 longitude.

Composite Heat Shields Validated

CHeSS was simulated to enter the Martian atmosphere at an altitude of 125.0 km with an inertial velocity of 7.4 km/sec, nominal gamma of -12.5 and a nominal angle-of-attack of -11.1. It was found that the first stage heat shield of the CHeSS experienced a peak stagnation point convective heat flux of 92.5 watts/cm^2 which is shown in Fig. 10. A negligible radiative heat flux is shown in Fig. 11. Fig. 12 compares the first stage convective heat flux to the values calculated for Pathfinder based upon the Braun, et.al relative entry conditions.[5] The peak value for the Pathfinder case is 113 watts/cm^2. The peak convective heat flux value for CHeSS is lower than the Pathfinder value but the integrated heat flux over time for CHeSS is almost double that for Pathfinder. However the Braun, et.al. simulation was one of many different simulations performed for Pathfinder. An unpublished simulation went as high as 132 watts/cm^2 with an integrated heat flux roughly 75% of the value for CHeSS. Fig. 13 shows a peak deceleration of 7.5 g. By comparison, Pathfinder's predicted peak deceleration is 28 g. Based upon these comparisons, a thickened Pathfinder aeroshell using SLA-561V Thermal Protection System (TPS) material as used in the Pathfinder mission maybe usable for the CHeSS first stage with the weight difference taken from the instrument-controller ring. Should it be necessary that the TPS material thickness be significantly increased then there are alternative materials such as Silicone Impregnated Reusable Ceramic Ablator (SIRCA) that could tolerate a higher integrated heat flux than SLA-561V.[18]

The peak convective heat flux of the second stage heat shield (AMaRV) is shown in Fig 10 to be 153 watts/cm^2. This heat flux is high enough to cause TPS material recession if SLA-561V were used. Since the second stage is a precision maneuvering vehicle, changes in the aerodynamic geometry are undesired. However reinforced carbon-carbon TPS material should tolerate heat fluxes predicted for the AMaRV second stage without significant recession. Fig. 14 shows the stagnation point heat flux for CHeSS never exceeded 2500 deg.K. The sublimation temperature for carbon graphite is 3640 deg.K. which is less than the peak heat flux. However Martian atmospheric carbon dioxide would dissociate to yield atomic oxygen and this could cause burning. This burning is probably less than what AMaRV was designed for in its terrestrial application where carbon-carbon TPS was used. The ablation rate due to carbon-carbon burning in the Martian atmosphere is an area open for study.

Figures 15, 16 & 17 show the nominal CHeSS trajectory of inertial entry angle -12.5 deg along with the limiting error trajectories of entry angle -12.0 and -13.0 deg. A first stage angle of attack of -8.55 deg. is used with the entry angle of -12.0 deg. The nominal first stage angle of attack of -11.1 is used with the nominal entry angle of -12.5 deg. and -13.3 is the first stage angle of attack for the entry angle of -13.0. This sets the second skip shown in Fig. 15 to almost the same location. To remove the remaining error, the AMaRV second stage is set to an angle of attack of 12.35 deg. for the original entry angle of -12.0 deg., 10.0 deg. for the nominal entry angle of -12.5, and 9.64 deg. for the entry angle of -13.0. The final impact location is within 20 km for all three trajectories as shown in Fig. 17. It should be emphasized that assuming constant angle of attack is a naive control algorithm for such a highly maneuverable vehicle like AMaRV. An actual mission would involve a more sophisticated control law using an inertial measurement unit, radar altimeter (when the plasma sheath allowed) and local horizon sensor. What was demonstrated by this naive example was the ability to perform accurate navigation while not exceeding the limits of a realistic thermal protection system.

The effectiveness of the composite heat shield concept is made obvious when compared to an unmodified AMaRV entering the Martian atmosphere from Pathfinder entry conditions. For that case it can be seen in Fig. 18 that AMaRV experienced a peak stagnation point heat flux of 3856 watts/cm^2. Fig. 19 shows that after reaching a minimum altitude of 80 meters (at Mach thirty!), AMaRV did not decelerate sufficiently to be captured by Mars but left the vicinity of Mars on a hyperbolic escape trajectory. The atmosphere of Mars is simply too thin to aerocapture a low drag entry vehicle.

For the opposite scenario of CHeSS reentering the Martian atmosphere on its first stage heat shield, it was found that CHeSS had too much drag to skip off of the atmosphere a second time and impacted the Martian surface.

Potential Applications for CHeSS

CHeSS was invented to demonstrate a practical example of a composite heat shield. The design used unmodified flight proven hardware for components so that attention could be focused upon CHeSS as a total unit. It was an unexpected surprise that CHeSS had a total launch mass less than Pathfinder's and could easily fit within the payload faring of a Delta-II launch vehicle. CHeSS could actually be built and sent to Mars at a relatively low cost. This leads to an interesting line of speculation on what potential applications CHeSS could have in exploring Mars.

If it is assumed the stall angle-of-attack for AMaRV is 25 degrees then the stall speed near the Martian surface is 980 meters/sec which corresponds to Mach 4.19. The dynamic pressure experienced at stall is 7251 Pascals. Viking-I deployed its parachutes at Mach 1.1 with a dynamic pressure of 321 Pascals. Viking's design limit for successful parachute deployment was Mach 2.1 with a dynamic pressure of 412 Pascals.[19] Pathfinder was designed to deploy its parachute at Mach 1.9 with a dynamic pressure of 660 Pascals. It is obvious the high stall speed of AMaRV prohibits use of a Viking/Pathfinder type parachute system. However there have been supersonic deceleration systems successfully used to recover military reentry vehicles.[20] For example, the IRS-II recovery system deployed a 0.5 meter diameter parachute at Mach 1.9 with a dynamic pressure of 215 kiloPascals. Although the Mach number was less than half of the second stage AMaRV's, the dynamic pressure was almost thirty times greater. A 0.5 meter parachute could not significantly decelerate a 470 kg AMaRV before impacting the Martian surface. However the AMaRV could eject a small payload that decelerated to a survivable speed using the parachute. One can imagine a mission scenario where the AMaRV jettisoned a payload at minimum altitude for every skip. Thus one AMaRV could precisely eject six payloads separated by thousands of kilometers on the Martian surface.

Another mission scenario could involve remote sensing while flying at high speed but low altitude. The AMaRV could have an ultraviolet laser shining a beam along its ground track while an on-board telescope observed surface florescence. Mars has a very weak magnetic field. It might be possible for an AMaRV cruising at low altitude to make magnetometer measurements thus providing better understanding of the Martian magnetic field and perhaps detecting the location of metallic ore bodies through magnetic anomaly.

Perhaps the most interesting potential application for CHeSS is its use for a low cost Mars sample return. The main difficulty with a conventional sample return is the spacecraft must land on the surface of Mars giving up all of its kinetic energy. The most ubiquitous substance on the surface of Mars is fine dust. The dust storms on Mars can be spectacular. Papadopoulos, et. al. found that dust suspended in the Martian atmosphere could cause erosion of a spacecraft's heat shield.[21] After a dust storm, there remains a significant amount of dust suspended in the atmosphere. A second stage AMaRV could plunge down into the atmosphere, collect a small quantity of suspended dust, skip out of the atmosphere and then insert the dust sample into a stable orbit using a small solid rocket. Another spacecraft (perhaps using the first stage CHeSS heat shield) could then collect the dust sample from Mars orbit and return it to Earth. The Stardust spacecraft which is currently being designed uses aerogel to collect coma dust impacting at 6.1 km/sec. The AMaRV second stage could have along its axis of symmetry a shock tube filled with high pressure xenon gas. One end of this shock tube has a nozzle and burst diaphragm located just behind the AMaRV nose cap. The other end of the shock tube contains a quantity of aerogel. To collect atmospheric dust, the AMaRV descends to 10 km altitude and cruises at about 3 km/sec. The burst diaphragm is ruptured, blowing the nose cap off and jetting out xenon gas through a choked nozzle. Suspended dust particles pass through the shock wave, shock layer, contact discontinuity and into the cold expanding xenon gas. The dust particles while decelerating in the dense xenon gas pass through the shock tube's nozzle into the shock tube and impact the aerogel at the shock tube's other end. The xenon gas not only decelerates the particles but also shields the aerogel from the hot gases of the shock layer. Before the burst diaphragm expansion wave reaches the aerogel, a gate valve is closed by a pyrotechnic device sealing the shocktube with its collected dust. The outer surface of the gate valve is made of TPS material since it acts as the new nose cap. The AMaRV then does a pitch-up maneuver, exits the atmosphere and inserts the shock tube into a stable orbit using a solid rocket motor. If the dust collection time was one millisecond and shocktube collection radius one centimeter then 942 cm^3 of Martian atmosphere would be sampled for dust. Papadopoulos, et. al. assumed a dust to atmosphere mass ratio of 10^{-4}. If dust collection occurred at an altitude of 10 kilometers then the estimated total mass of dust collected would be 0.66 milligrams.

Concluding Remarks

A Martian atmospheric entry program was written and validated against analytic solutions and Viking-I flight data. This program demonstrated feasibility for an example composite heat shielded spacecraft using a Viking/Pathfinder aeroshell first stage and AMaRV second stage constructed with conventional TPS materials. It was argued that a control strategy using center-of-mass variation was preferable. It was shown that this spacecraft offered interesting opportunities for exploring Mars including the possibility of returning a Mars dust sample.

Acknowledgement

The author gratefully acknowledges the support of the Reacting Flow Environments Branch at NASA Ames Research Center through Contract NAS2-14031 to Eloret.

References

[1] Tauber, M.E. and Seiff, A., "Optimization of Heating of Conical Bodies Making Lifting Hyperbolic Entries into the Atmospheres of Earth and Mars," *Proceedings of the AIAA Entry Technology Conference*, Williamsburg, VA, October 1964.

[2] Repic, E.M., Boobar, M.G. and Chapel, F.G., "Aerobraking as a Potential Planetary Capture Mode," *Journal of Spacecraft and Rockets*, Vol. 5, No. 8, August 1968, pp. 921-926.

[3] Tauber, M.E., Bowles, J.V. and Yang, L., "Use of Atmospheric Braking During Mars Missions," *Journal of Spacecraft and Rockets*, Vol. 27, No. 5, Sept-Oct 1990, pp. 514-521.

[4] Allen, H.J. and Eggers, A.J., "A Study of Motion and Aerodynamic Heating of Ballistic Missiles Entering the Earth's Atmosphere at High Supersonic Speeds," NACA Report 1381, 1958.

[5] Braun, R.D., Powell, RW., Engelund, W.C., Gnoffo, P.A., Weilmuenster, K.J. and Mitcheltree, R.A., "Six Degree-of-Freedom Atmospheric Entry Analysis for the Mars Pathfinder Mission," AIAA Paper 95-0456, January 1995.

[6] Tauber, M.E., Chargin, M., Henline, W., Chiu, A., Yang, L., Hamm, Jr. , K.R. and Miura, H. "Aerobrake Design Studies for Manned Mars Missions," AIAA Paper 91-1344, June 1991.

[7] Cruz, M.I., "Technology Requirements for a Generic Aerocapture System," AIAA Paper 80-1493, July 1980.

[8] Tauber, M.E. & Sutton, K., "Stagnation-Point Radiative Heating Relations for Earth and Mars Entries," *Journal of Spacecraft and Rockets*, Vol. 28, No. 1, Jan-Feb 1991, pp. 40-42.

[9] Chapman. D.R., "An Approximate Analytic Method for Studying Entry into Planetary Atmospheres," NACA TN-4276 (superseded by NASA Report R-11), May 1958.

[10] Chapman. D.R., "An Analysis of the Corridor and Guidance Requirements for Supercircular Entry into Planetary Atmospheres," NASA Report R-55, 1960.

[11] Personal communication, John Copper, McDonnell Douglas Corp., September 1995.

[12] Braun, R.D., Powell, R.W. & Hartung, L.C., "Effect of Interplanetary Trajectory Options on a Manned Mars Aerobrake Configuration," NASA Technical Paper 3019, August 1990, pg. 30.

[13] Regan, F.J. & Anandakrishnan, S.M., *Dynamics of Atmospheric Re-Entry*, AIAA, Washington, 1993, pg. 142.

[14] Vinh, N.X., *Optimal Trajectories in Atmospheric Flight,* Elsevier, 1981.

[15] Seiff, A., "Post-Viking Models for the Structure of the Summer Atmosphere of Mars," *Adv. Space Res.*, Vol. 2, 1982 pp. 3-17,.

[16] Marvin, J.G. & Deiwert, G.S., "Convective Heat Transfer in Planetary Gases," NASA TR R-224, July 1965.

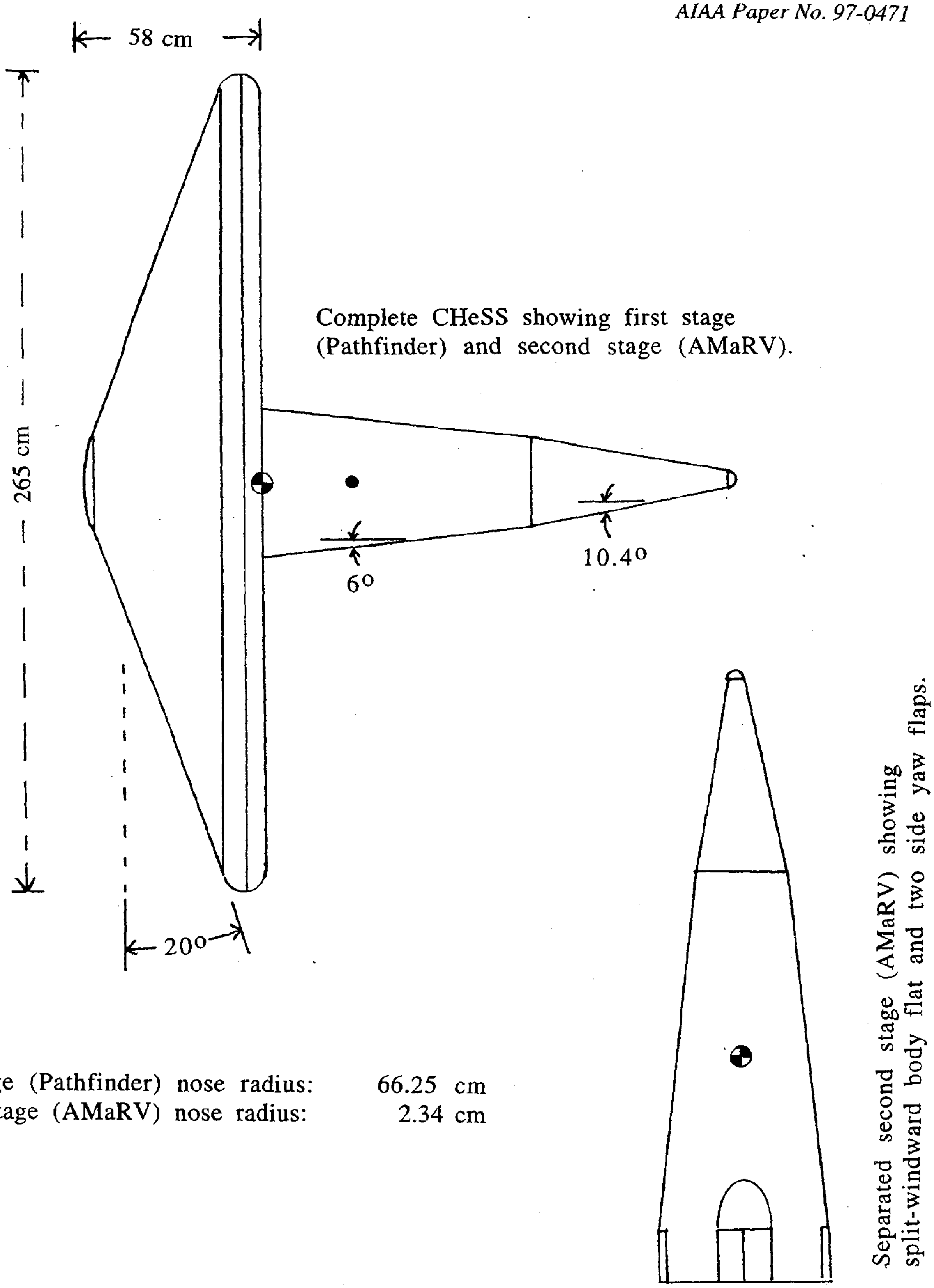
[17] Ingoldby, R.N., et. al., "Entry Data Analysis for Viking Landers 1 and 2," NASA Contractors Report CR-159388 (Martin Marietta Corp. Report TN-3770218) , November 1976.

[18] Wercinski, P.F., "Mars Aerocapture Analysis for the MESUR/Mars-Pathfinder Aeroshell in Low L/D Configurations," AIAA Paper 95-3495, August 1995.

[19] Cooley, C.G., et. al., "Viking 75 Project, Viking Lander System Primary Mission Performance Report," NASA Contractors Report CR-145148, April 1977, pg. V-2.

[20] Rigali, D.J., Sterk, M.W. and Randmaa, J., "Re-Entry Vehicle Soft-Recovery Techniques," AIAA Paper 80-1455, July 1980.

[21] Papadopoulos, P., Tauber, M.E. & Chang, I.-D., "Heatshield Erosion in a Dusty Martian Atmosphere," *Journal of Spacecraft and Rockets*, Vol. 30, No. 2, March-April 1993, pp. 140-151.



First stage (Pathfinder) nose radius: 66.25 cm
Second stage (AMaRV) nose radius: 2.34 cm

Fig. 1 Composite Heat Shielded Spacecraft (CHeSS) with second stage shown separately.

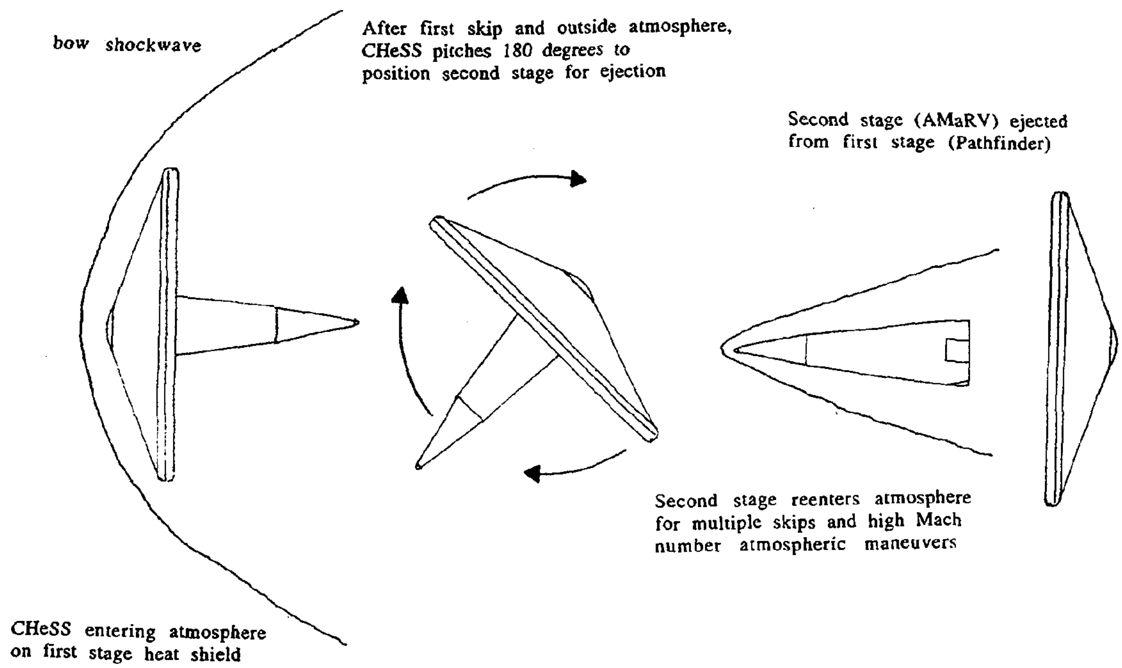


Fig. 2 Staging of CHeSS.

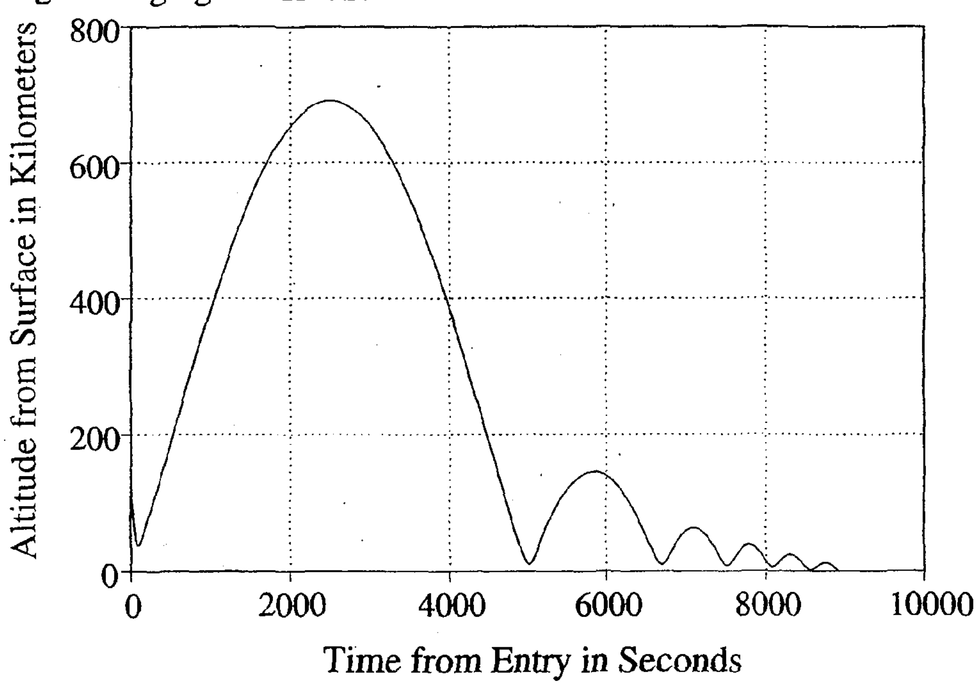


Fig. 3 CHeSS Altitude versus Time.

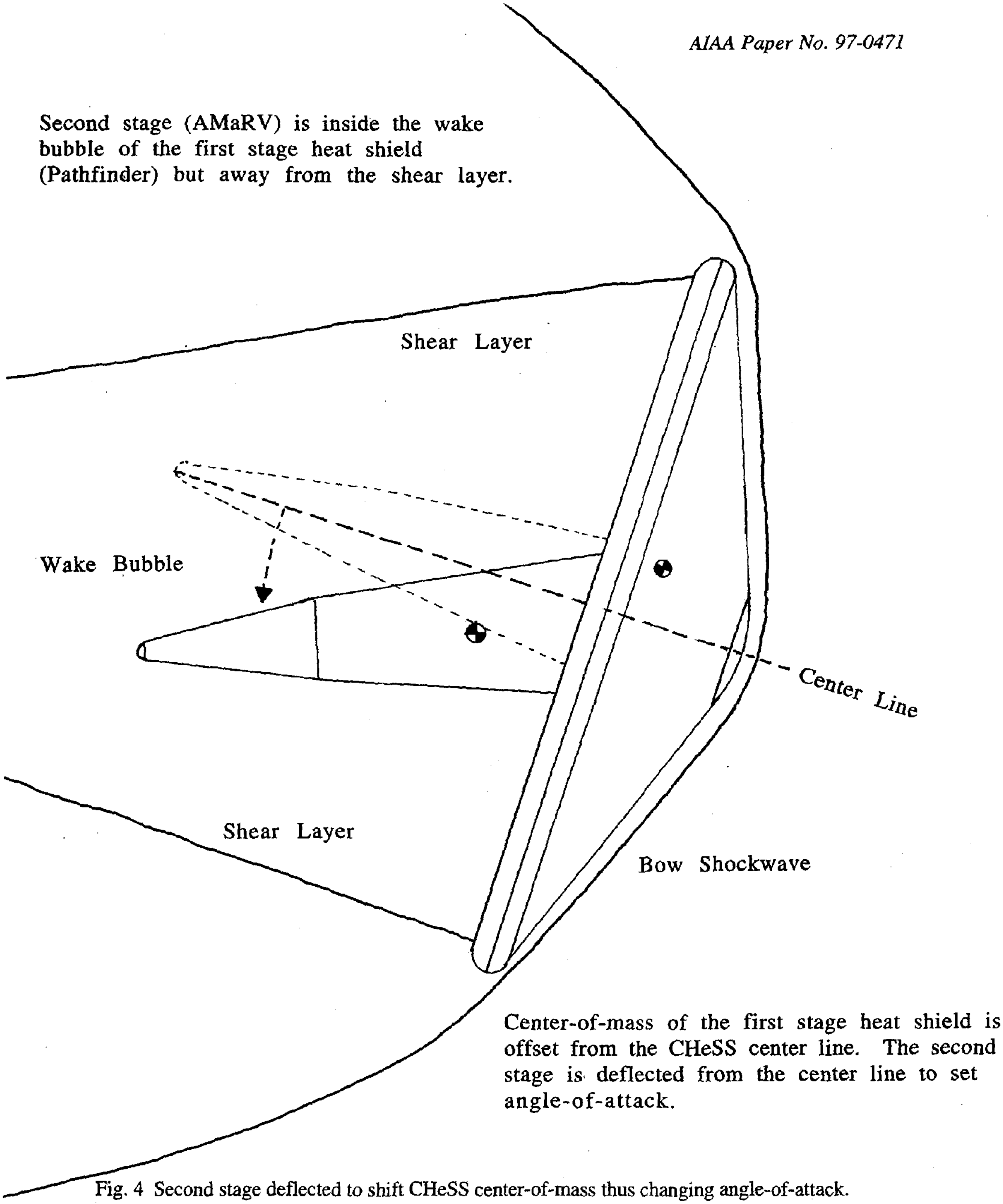


Fig. 4 Second stage deflected to shift CHeSS center-of-mass thus changing angle-of-attack.

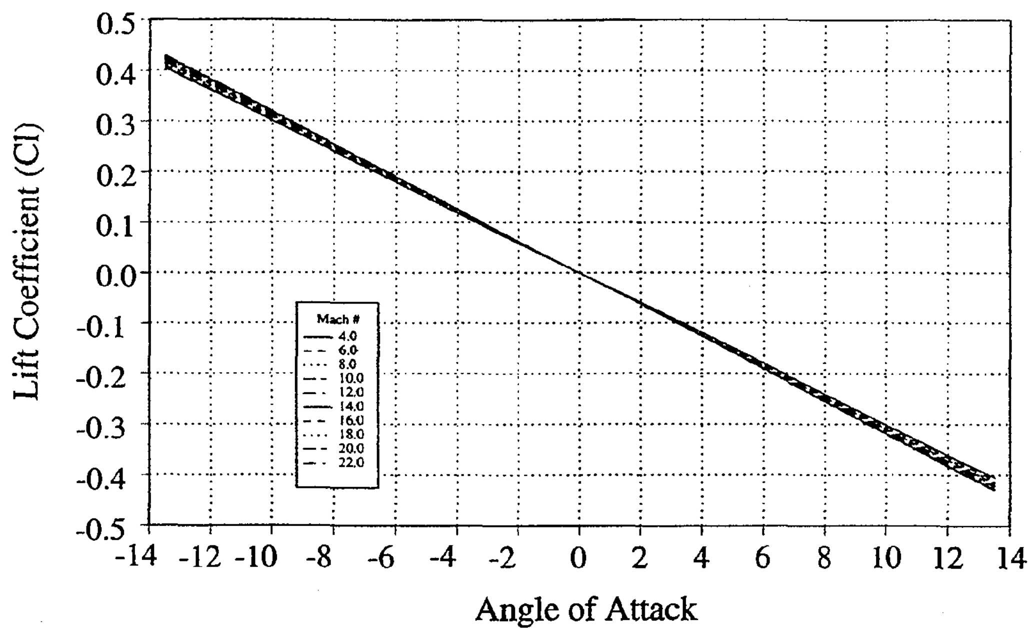


Fig. 5 Viking Lift Coefficient vs. Angle of Attack.

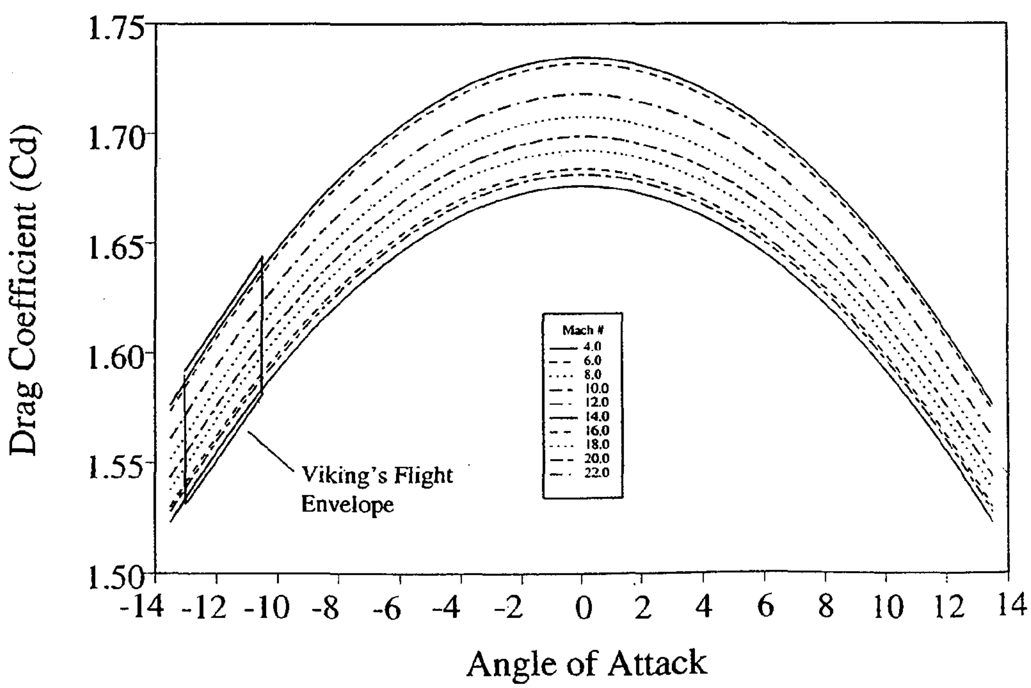


Fig. 6 Viking Drag Coefficient vs. Angle of Attack.

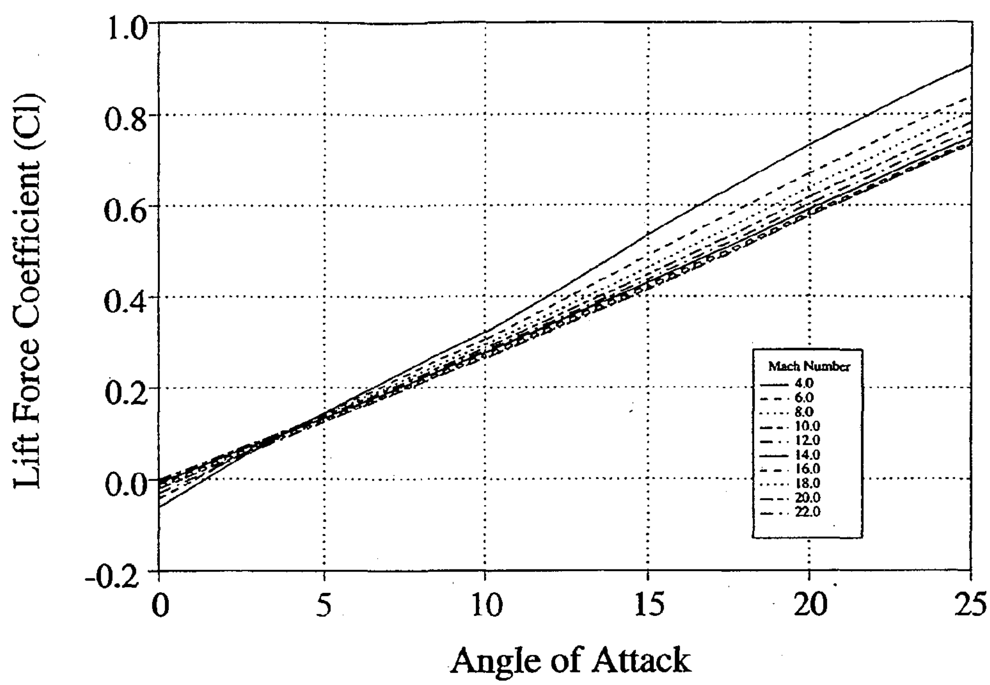


Fig. 7 AMaRV Lift Coefficient vs. Angle of Attack.

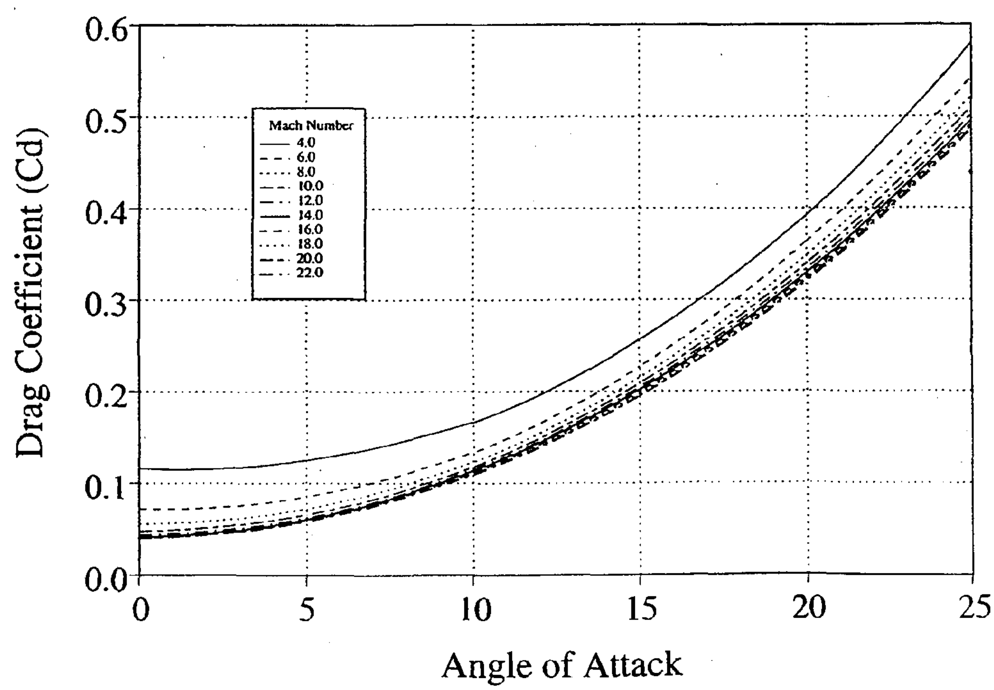


Fig. 8 AMaRV Drag Coefficient vs. Angle of Attack.

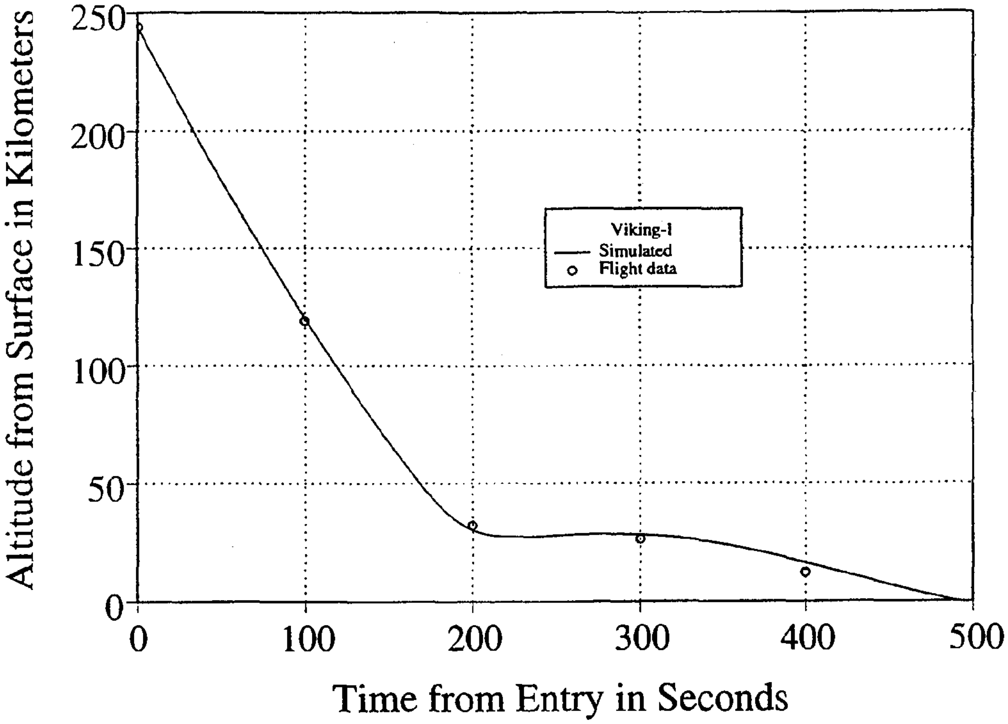


Fig. 9 Simulated Viking-I Altitude versus Time compared to Flight Data.

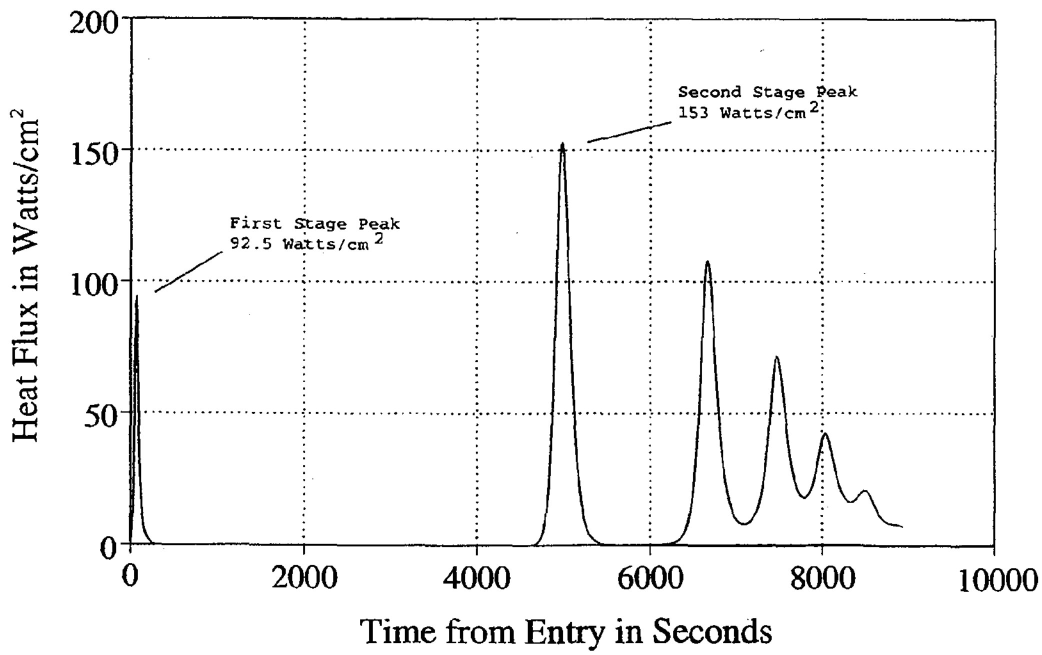


Fig. 10 CHeSS Heat Flux versus Time.

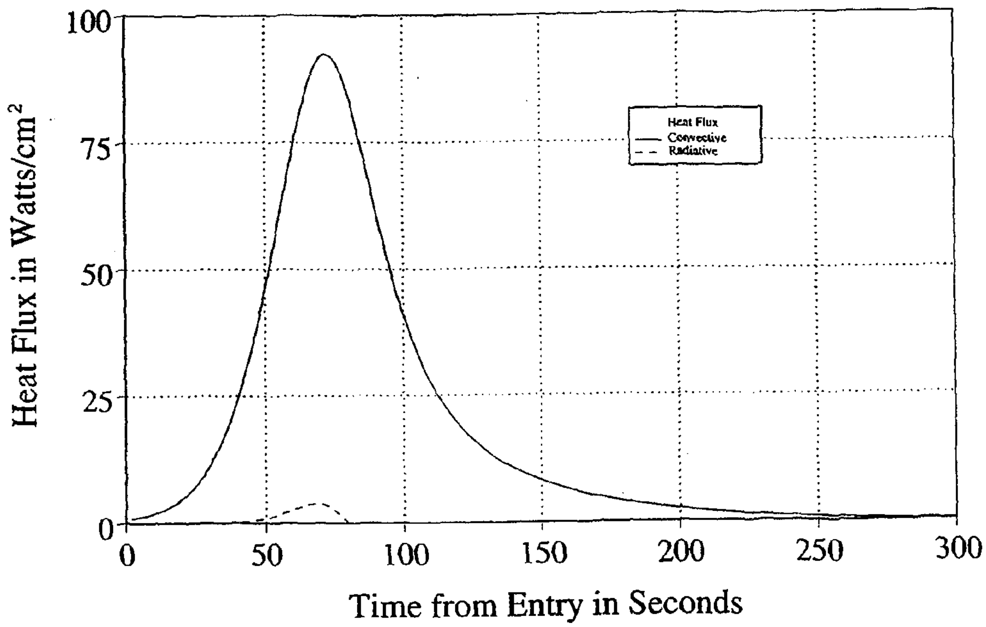


Fig. 11 First Stage of CHeSS Heat Flux versus Time.

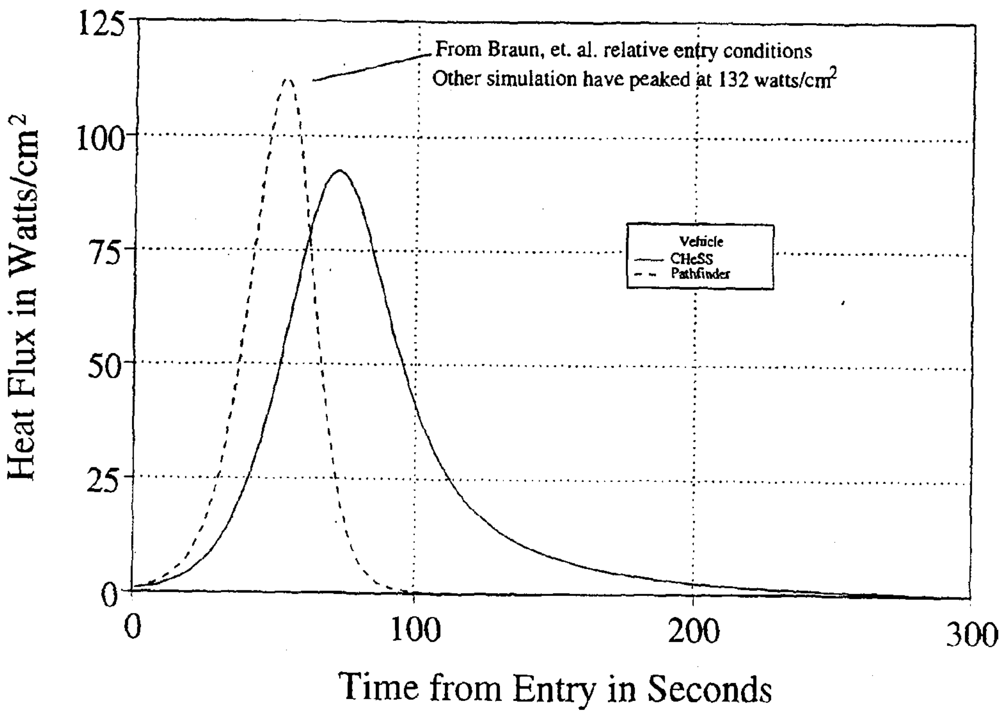


Fig. 12 Heat flux versus time for CHeSS compared to Pathfinder.

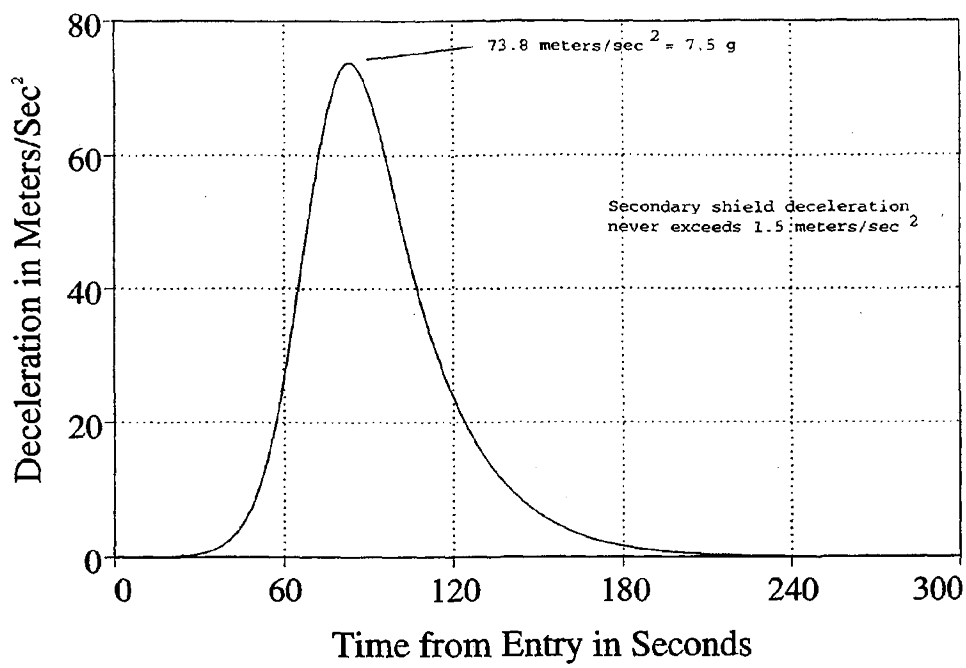


Fig. 13 First Stage of CHeSS Deceleration versus Time.

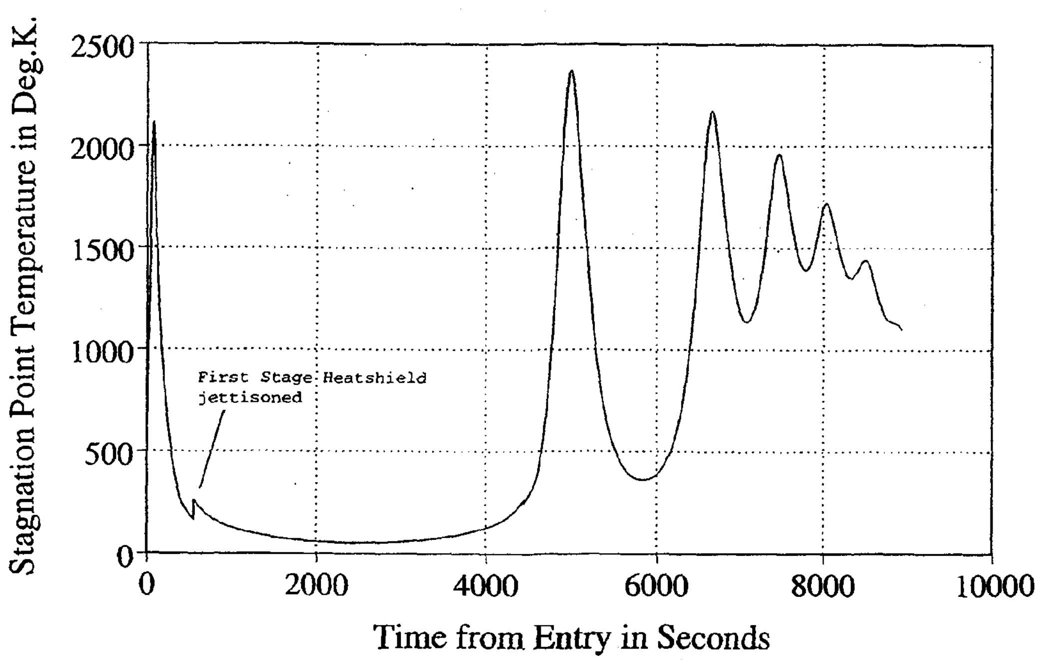


Fig. 14 CHeSS Stagnation Point Temperature versus Time.

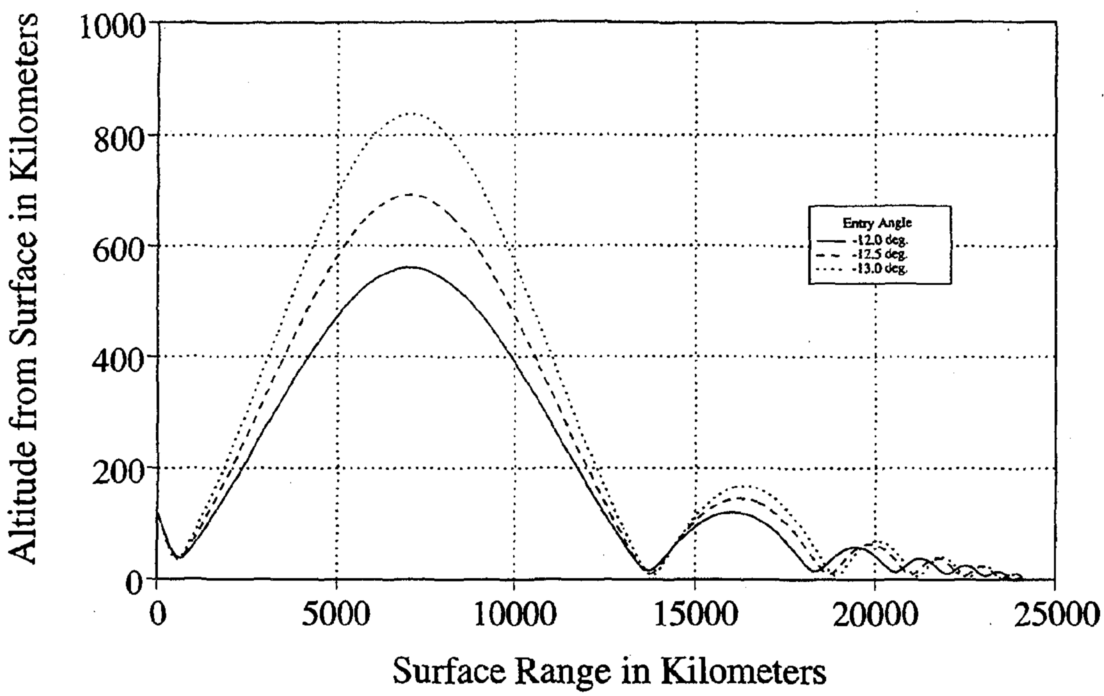


Fig. 15 CHeSS Altitude versus Range for Nominal and Error Entry Angles.

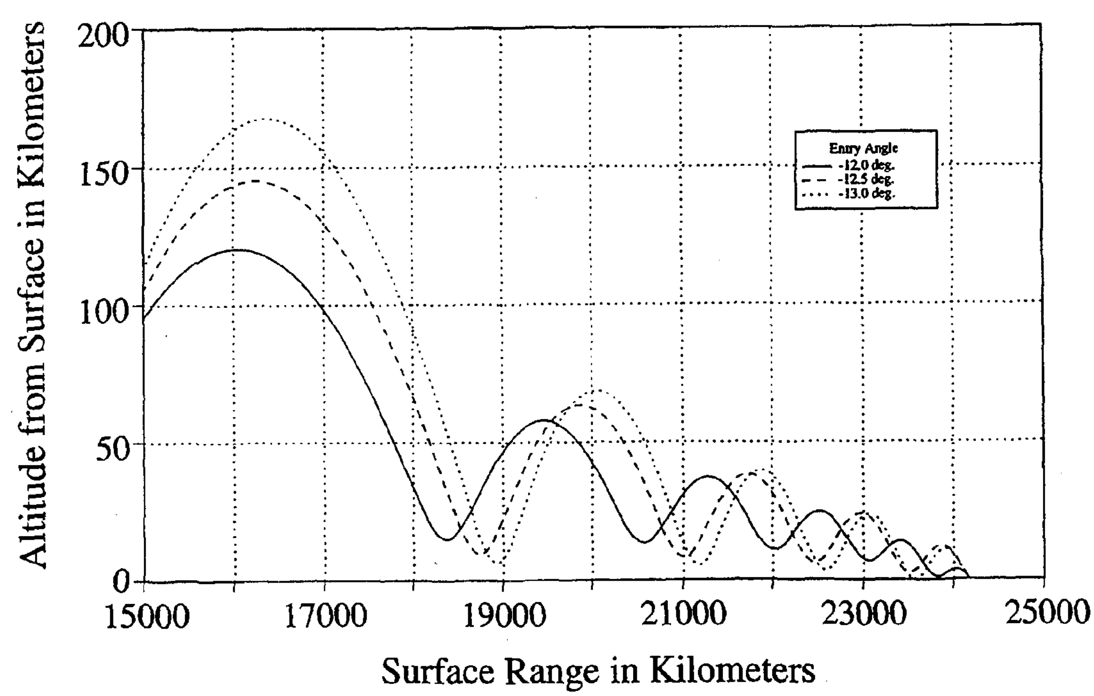


Fig. 16 CHeSS Altitude versus Range Showing Second Stage Atmospheric Skips for Nominal and Error Entry Angles.

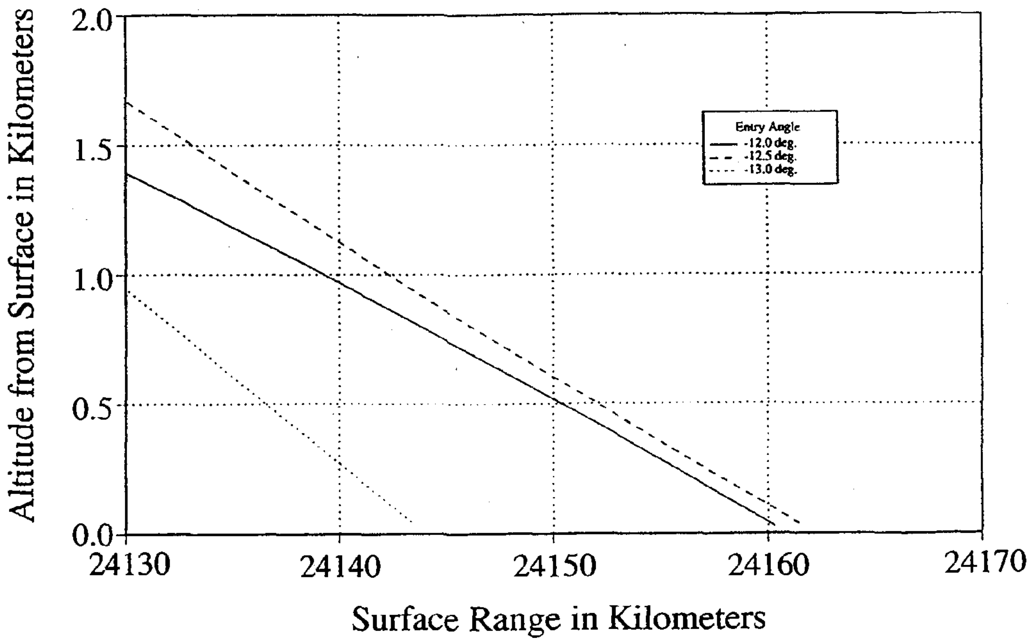


Fig. 17 CHeSS Altitude versus Range Showing Second Stage Impact on Martian Surface.

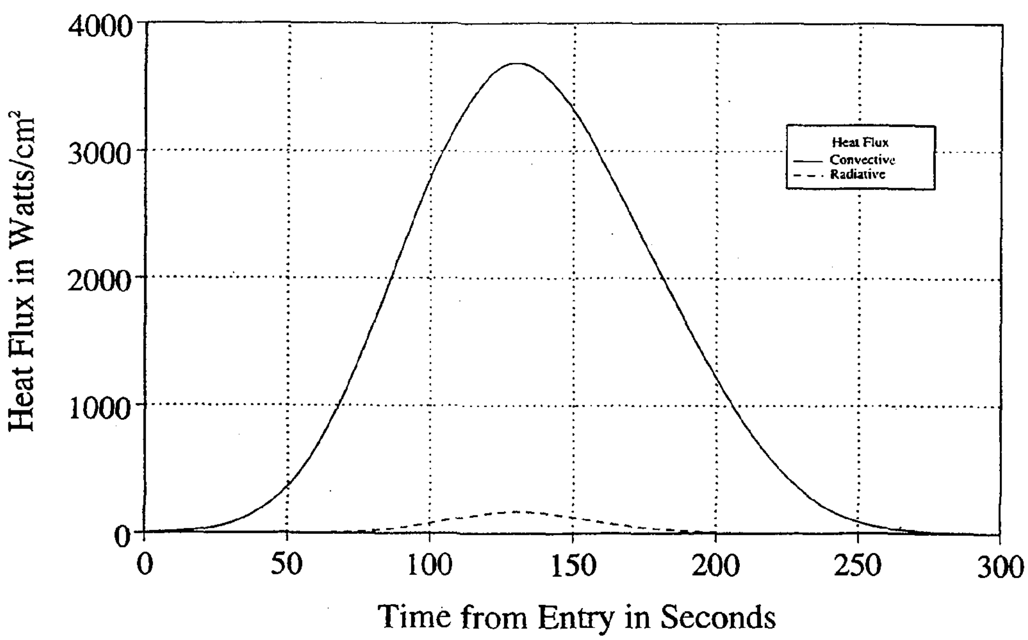


Fig. 18 AMaRV Direct Entry Heat Flux versus Time.

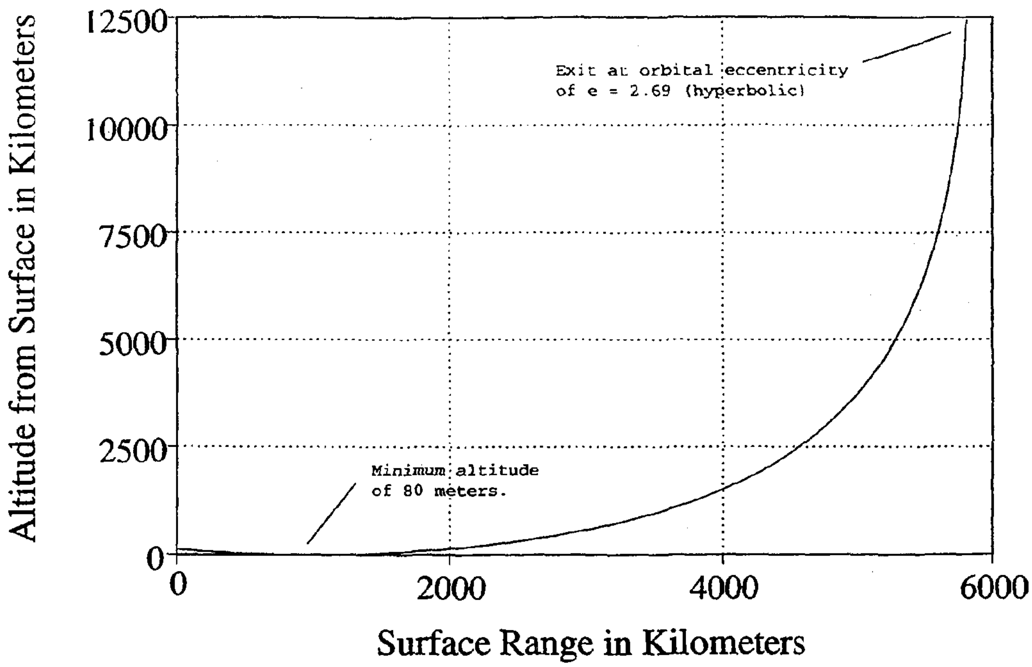


Fig. 19 AMaRV Direct Entry Altitude versus Range Showing Escape from Martian Vicinity.